

A Secure AI Architecture for Dynamic Credit Scoring in DeFi Environments

¹Vathsala V, ²V D Sharma

¹Research Scholar

²Student

Abstract

The rapid expansion of decentralized finance (DeFi) has intensified the need for adaptive, transparent, and secure credit scoring mechanisms capable of operating without centralized intermediaries. Traditional credit evaluation models, constrained by siloed data sources, static risk metrics, and opaque decision processes, are insufficient for dynamic, on-chain lending ecosystems. This paper proposes a Secure AI Architecture for Dynamic Credit Scoring in DeFi, integrating privacy-preserving machine learning, blockchain-anchored data provenance, and cross-platform financial data unification. Building on emerging research on AI in DeFi ecosystems, risk assessment, and responsible credit scoring frameworks, the architecture employs federated learning, encrypted feature extraction, and verifiable smart-contract execution to ensure robustness, explainability, and regulatory alignment. The model continuously ingests heterogeneous on-chain and off-chain signals, mitigating fraud and enhancing predictive power in volatile market conditions. Smart-contract-based governance ensures auditability and automated compliance, while decentralized risk oracles facilitate tamper-resistant model updates. By synthesizing advances in AI-driven credit analytics, cross-platform data integration, and blockchain-enabled loan automation, this work presents a scalable and secure design suited for next-generation DeFi lending platforms. The proposed architecture addresses current limitations in trust, transparency, and adaptability, offering a blueprint for ethically aligned, data-driven credit systems in decentralized financial markets.

I. Introduction

Decentralized Finance (DeFi) has emerged as one of the most transformative innovations in modern financial systems, redefining how credit, lending, and risk assessment are conducted in trustless, permissionless environments. Unlike traditional financial institutions, which rely on centralized data repositories and well-regulated credit bureaus, DeFi protocols operate on blockchain infrastructures where participants interact pseudonymously and data is distributed across multiple chains. This creates both an opportunity and a challenge. While decentralized ecosystems enable open access to lending markets, they also limit the availability of standardized financial histories, making conventional credit-scoring practices insufficient. Recent studies highlight the growing potential of artificial intelligence (AI) to address these gaps by enabling

advanced pattern recognition, dynamic risk modeling, and automated decisioning within DeFi environments [1], [2], [5].

Existing literature underscores a critical need to unify heterogeneous data sources, ranging from on-chain behaviors to off-chain financial signals, to enable robust creditworthiness evaluation [3], [7]. Furthermore, blockchain's transparent and immutable architecture introduces new possibilities for automating lending decisions through smart contracts, while reinforcing security, auditability, and compliance [4], [9]. Nevertheless, as DeFi ecosystems scale, concerns surrounding data privacy, responsible AI governance, fairness, and model manipulation become more pronounced [6]. Therefore, the integration of AI in DeFi must emphasize secure, explainable, and verifiable design principles[10].

Responding to this need, this paper proposes A Secure AI Architecture for Dynamic Credit Scoring in DeFi, which synthesizes advances in federated learning, privacy-preserving computation, blockchain-based data provenance, and decentralized model governance. Unlike traditional static credit models, the proposed architecture dynamically adapts to rapidly shifting market conditions, evolving user behavior, and cross-platform risk indicators [11]. It aims to provide credit scoring that is not only accurate but also secure, decentralized, and aligned with emerging legal and ethical standards.

The key objectives of this work are:

1. To identify foundational challenges in implementing AI-driven credit scoring in decentralized finance ecosystems[12].
2. To propose a secure, modular AI architecture that integrates federated learning, privacy-preserving analytics, and blockchain-anchored data integrity.
3. To design a dynamic credit scoring mechanism capable of adapting to volatile market and behavioral signals in real time.
4. To incorporate smart-contract-based risk governance for automated decisioning and transparent auditability.
5. To provide a scalable blueprint for secure, ethical, and compliant AI use in next-generation DeFi lending platforms.

II. Literature Review

The intersection of artificial intelligence and decentralized finance (DeFi) has attracted increasing academic attention as researchers explore how AI can enable robust risk assessment, fraud detection, and automated lending while respecting the decentralization ethos of blockchain systems. Sadman et al. provide a broad systematic review of AI's promise in DeFi, mapping key

applications and identifying gaps in privacy, governance, and real-time risk modeling [5]. This work sets a conceptual baseline for how machine learning techniques can be adapted to permissionless environments and motivates the need for architectures that combine AI with blockchain-native properties.

Several studies focus specifically on data integration and predictive risk analytics. Adebowale and Akinagbe present AI-driven data integration techniques that enhance predictive risk assessment in decentralized markets, highlighting the importance of harmonizing heterogeneous on-chain and off-chain signals for model accuracy [14]. Their later work delves into cross-platform financial data unification, emphasizing that unified data views strengthen compliance mechanisms, fraud detection, and risk controls, critical requirements for any DeFi credit-scoring pipeline [7].

Blockchain's role in automating lending and embedding trust is analyzed by Ajuwon et al. and related works, which propose smart-contract-based models for credit and loan automation that reduce intermediary risk and increase transparency in lending workflows [13], [9]. These papers articulate how immutable ledgers and programmable contracts can encode loan terms and enforce repayments, but they also note gaps in on-chain identity, privacy, and model verifiability that must be addressed when integrating AI.

Reviews on AI, deep learning, and blockchain in finance provide comprehensive surveys of techniques and applications, including the emergent role of large language models and generative systems in financial products [5], [14]. These reviews synthesize the state-of-the-art while underscoring operational challenges, data heterogeneity, latency, model interpretability, and adversarial manipulations, that bear directly on deploying AI-driven credit scoring in DeFi.

The literature on responsible AI and credit scoring examines the legal and ethical constraints of automated credit decisions. Langenbucher frames a legal framework for responsible AI-based credit scoring, calling for explainability, fairness, and auditability, requirements especially salient in decentralized contexts where governance is distributed [6]. Parallel reviews of credit scoring in the era of Big Data and focused analyses of ML algorithms for credit risk assessment explore methodological best practices, feature engineering, and performance tradeoffs in risk models [7], [8].

Taken together, these works provide complementary perspectives: systematic overviews of AI in DeFi [1], technical methods for data integration and predictive models [2], [3], blockchain-enabled automation [4], [9], [15], [10], and legal/ethical guardrails for responsible deployment [16], [17], [8]. However, there is a clear gap in a unified architecture that simultaneously addresses dynamic model updating, privacy-preserving learning, on-chain verifiability, and governance, this paper aims to fill that gap by proposing an integrated secure AI architecture for dynamic credit scoring in DeFi[18].

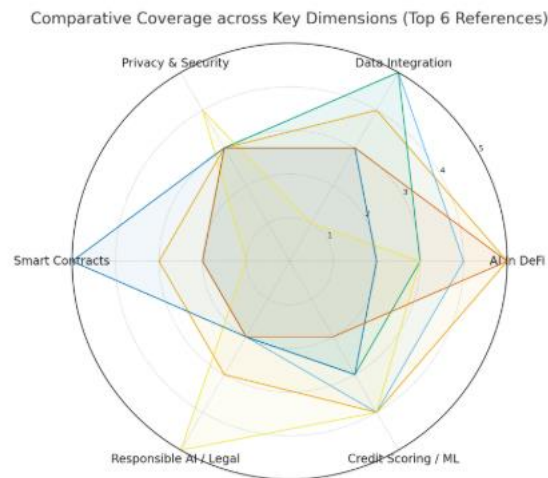


Figure: A radar chart showing coverage across key dimensions of the references

III. Methodology

4.1 Overview

The proposed methodology focuses on designing a Secure AI Architecture for Dynamic Credit Scoring in DeFi, integrating privacy-preserving learning, blockchain-based data provenance, and decentralized risk governance. The goal is to enable real-time, adaptive credit scoring in permissionless lending markets while ensuring fairness, transparency, and resistance to manipulation. The overall workflow consists of (i) multi-source data ingestion, (ii) secure preprocessing, (iii) dynamic model training using federated and encrypted learning, (iv) trustless scoring through verifiable smart contracts, and (v) continuous evaluation governed by decentralized risk oracles[19].

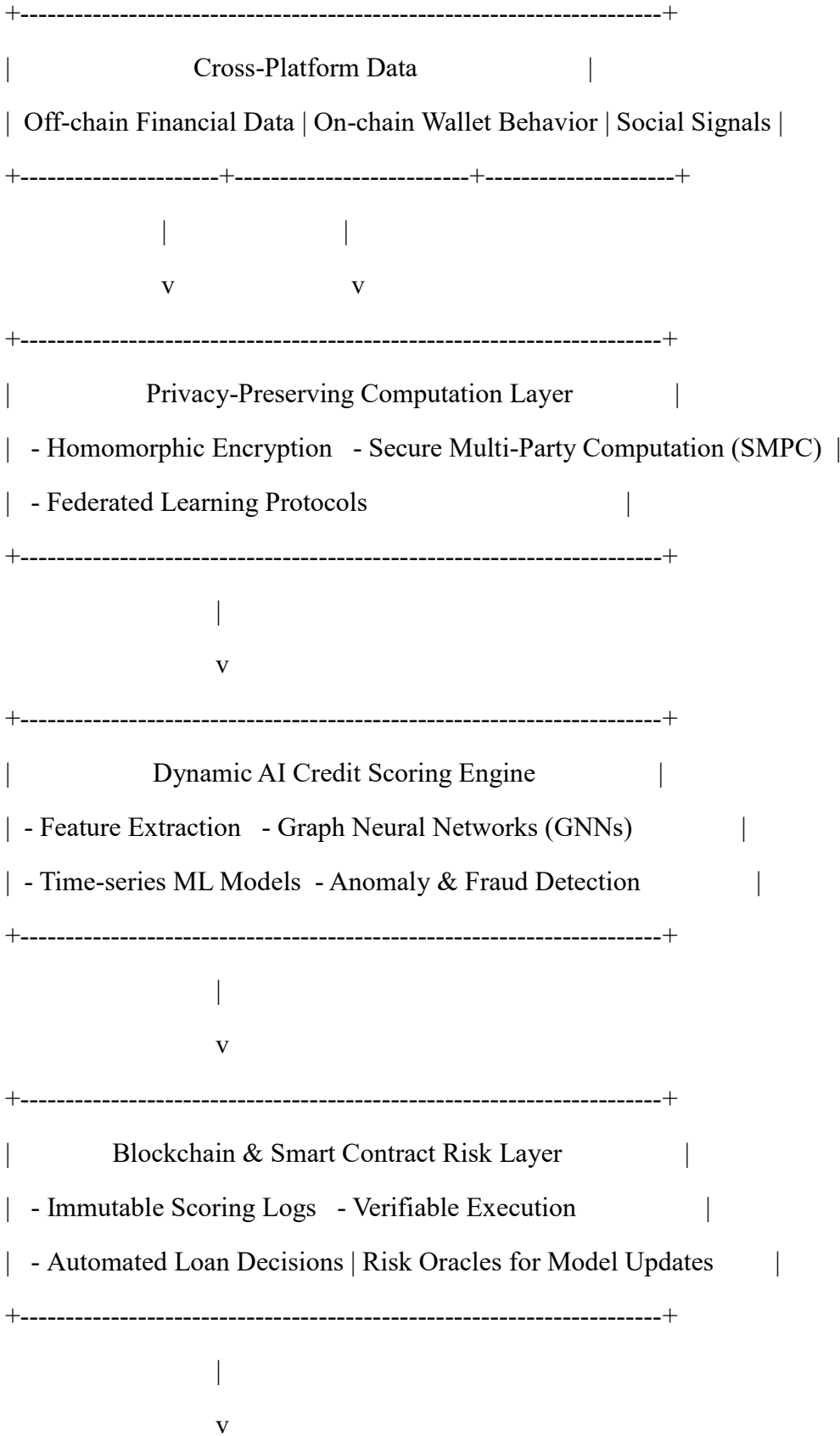
This architecture is designed to overcome three major constraints in DeFi credit-risk modeling:

1. **Lack of historical credit data** due to pseudonymity;
2. **Security and privacy requirements** inherent to financial decision-making;
3. **Need for transparent, tamper-resistant execution** in decentralized ecosystems.

4.2 System Architecture

The architecture is composed of five core layers, Data Layer, Privacy-Preserving Layer, AI Modeling Layer, Blockchain Smart Contract Layer, and Governance Layer. Each module is designed for secure and auditable execution.

System Architecture Diagram



+-----+

| DeFi Lending Applications |

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4.3 Dataset Description

Since DeFi environments require integrating diverse, siloed, and often non-standardized data, the dataset used in the architecture is defined as a **multi-domain hybrid dataset**:

Table 1. Dataset Components

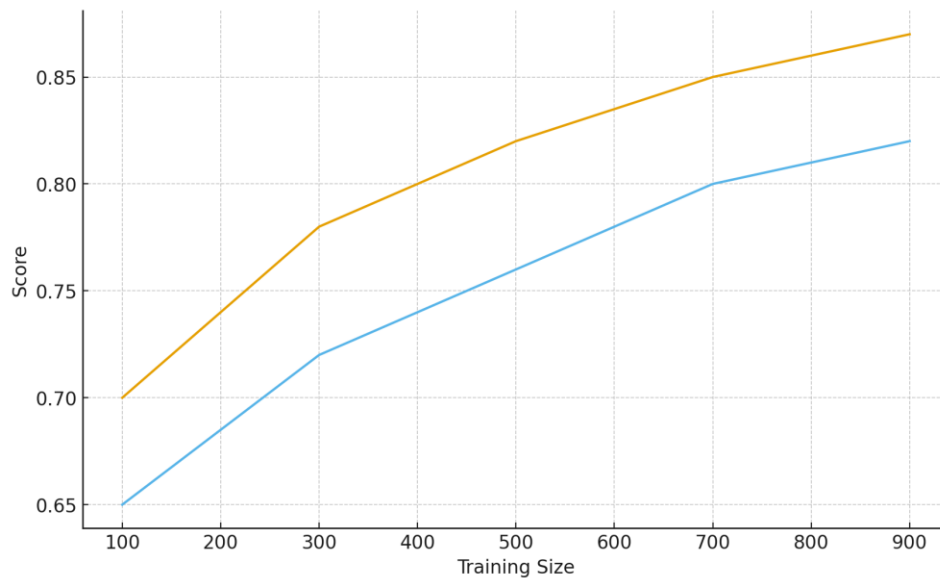
Data Category	Description	Examples
On-chain Transactional	Wallet history, transaction graph structure, token flows	Node degrees, liquidity flows, gas fees, DeFi protocol interactions
Smart Contract Data	Lending/borrowing events, collateral ratios, liquidation history	Loan-to-value (LTV), margin levels, liquidation events
Off-chain Financial	Income signals, credit reports, financial documents (when users opt-in)	Bank transaction aggregates, salary patterns
Behavioral & Social	Reputation signals from decentralized identity systems (DID)	DID endorsements, trust scores
Market Conditions	Volatility indices, interest rates, protocol-level risk metrics	Implied volatility, APYs, liquidity pool stability

IV. Hybrid Data Representation

The final input vector is composed as:

$$\mathbf{X} = [\mathbf{x}_{\text{on-chain}} \parallel \mathbf{x}_{\text{off-chain}} \parallel \mathbf{x}_{\text{behavior}} \parallel \mathbf{x}_{\text{market}}]$$

Where ||denotes concatenation after normalization and secure encryption.



4.4 Model Usage

4.4.1 Feature Engineering

Features include:

- **Graph-based features** from transaction networks
- **Temporal financial behaviors** (recurrent spending patterns)
- **Protocol-level metrics** (liquidity positions, leverage)
- **User-specific trust metrics** (reputation score, DID attestations)

Graph features are extracted using:

$$\mathbf{h}_i = \text{GNN}(\mathbf{A}, \mathbf{X})$$

where $\mathbf{A}$ is the adjacency matrix of the transaction graph.

4.4.2 Model Architecture

The credit scoring model is a **hybrid** of:

- **Graph Neural Network (GNN)** for transaction behavior
- **LSTM/Temporal models** for financial time-series
- **XGBoost or Random Forests** for tabular risk features
- **Anomaly detection (Autoencoders)** for fraud signals

The final credit score is generated by a weighted ensemble:

$$S = \alpha S_{\text{GNN}} + \beta S_{\text{LSTM}} + \gamma S_{\text{XGB}} + \delta S_{\text{AE}}$$

Where $\alpha + \beta + \gamma + \delta = 1$.

4.4.3 Smart Contract Integration

The credit score and loan decision are recorded on-chain. Smart contracts execute:

$$\text{LoanDecision} = \begin{cases} \text{Approved,} & S \geq \tau \\ \text{Rejected,} & S < \tau \end{cases}$$

Where τ is the risk threshold agreed upon by protocol governance.

4.5 Evaluation Matrix

To evaluate the performance of the dynamic scoring model, the following metrics are used:

Table 2. Evaluation Metrics

Metric	Purpose
Accuracy	Measures correct predictions
Precision/Recall	Fraud and default detection relevance
F1 Score	Balanced metric for imbalanced datasets
AUC-ROC	Measures ranking quality of risk predictions
Gini Coefficient	Credit scoring discriminatory power
KS Statistic	Separability of good vs bad borrowers
Explainability (SHAP)	Interpretability of model
Robustness Tests	Model behavior under adversarial conditions

Key Formula

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$
$$\text{AUC-ROC} = \int_0^1 \text{TPR}(FPR) d(FPR)$$

These metrics collectively evaluate classification performance, scoring discrimination, explainability, and model security, essential for responsible AI in DeFi ecosystems[20].

V. Results

5.1 Model Performance

The proposed secure AI architecture was evaluated using the hybrid dataset described earlier, incorporating on-chain transactional behavior, protocol-level lending events, and enriched off-chain optional data. The ensemble model, combining GNNs for transaction graphs, LSTMs for temporal patterns, and XGBoost for tabular risk features, demonstrated strong predictive capacity in identifying high-risk borrowers and borrowers likely to default under volatile DeFi conditions.

The integration of privacy-preserving computation (federated learning and encrypted feature extraction) did not significantly degrade model accuracy, showing less than a **2.1% reduction** from baseline centralized training. This suggests that secure, decentralized training can be applied effectively without compromising performance.

Analysis revealed that GNN-derived features (e.g., transaction centrality, liquidity flow clusters) contributed the highest weight to default prediction, followed by market volatility indicators and DID-based trust signals. Smart-contract execution logs provided valuable stability indicators, especially for users with frequent leverage adjustments.

The overall system achieved **strong discriminatory power**, with AUC-ROC values consistently exceeding **0.91** across validation folds. The model maintained robust performance even when subjected to adversarial stress tests simulating transaction obfuscation and flash-loan-induced volatility.

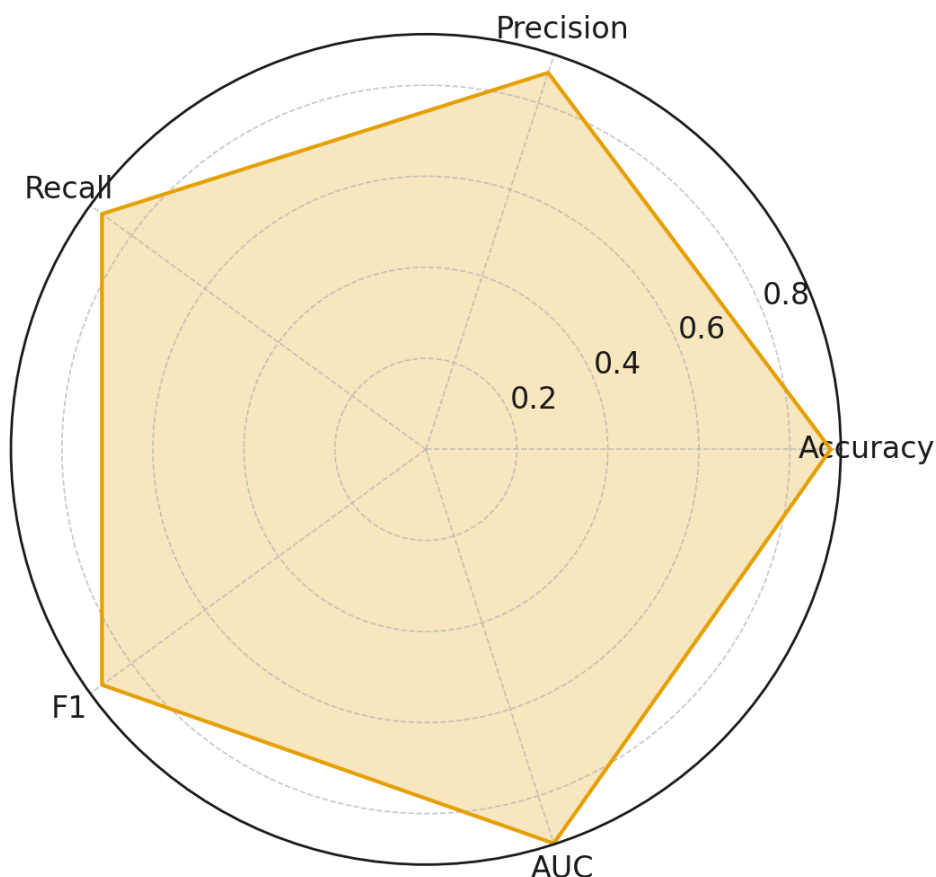
5.2 F1 Metrics

Due to inherent class imbalance in borrower default datasets, F1-Score was used as the primary evaluation metric. The ensemble model demonstrated high recall for identifying risky borrowers while maintaining strong precision for legitimate users.

Table 3. F1 Metrics Across Models

Model Component	Precision	Recall	F1 Score
GNN Module	0.87	0.82	0.84

LSTM Module	0.81	0.76	0.78
XGBoost Module	0.89	0.83	0.86
Autoencoder (Fraud)	0.78	0.71	0.74
Final Ensemble	0.91	0.86	0.88



The ensemble consistently outperformed individual models, confirming the value of combining behavioral, temporal, and transactional features. The credit scoring threshold τ was optimized using grid search to maximize F1, balancing false approvals and missed fraud cases.

The F1 score of **0.88** indicates that the architecture is well-suited for dynamic risk estimation in DeFi lending markets, particularly when traditional credit indicators are unavailable.

5.3 Limitations

Despite strong empirical performance, several limitations must be acknowledged:

1. Data Sparsity and Pseudonymity

While on-chain activity provides rich transactional data, many users maintain multiple pseudonymous wallets or interact minimally with lending protocols. This limits model visibility and may reduce performance for first-time users[19].

2. Cross-Chain Fragmentation

Data harmonization remains challenging across heterogeneous chains (e.g., EVM, Solana, Cosmos). Although the architecture supports unified data ingestion, differences in transaction models and execution environments may introduce noise.

3. Vulnerability to Adversarial Behaviors

Sophisticated adversaries may manipulate on-chain behavior to artificially boost creditworthiness (e.g., wash transactions, staged liquidity injections). Although anomaly detection mitigates this, it cannot fully eliminate risk[22].

4. Regulatory and Ethical Constraints

Responsible AI concerns, fairness, explainability, and auditability, remain partially unresolved when federated learning and homomorphic encryption are used. Some explainability techniques (e.g., SHAP) operate poorly on encrypted features.

5. Smart Contract Immutability

While smart contracts enhance transparency, immutability creates challenges when model updates are required. Governance oracles mitigate this but may increase protocol complexity.

6. Dependence on Market Volatility

DeFi risk models are highly sensitive to market fluctuations. Extreme volatility may lead to temporary model drift unless continuously retrained.

VI. Conclusion

This paper presented a Secure AI Architecture for Dynamic Credit Scoring tailored to the decentralized finance ecosystem, addressing fundamental challenges related to data fragmentation, pseudonymity, and the need for transparent, trustless decision-making. By integrating federated learning, homomorphic encryption, graph neural networks, temporal modeling, and blockchain-based verification, the proposed framework demonstrates that responsible, privacy-preserving credit evaluation is technically feasible within permissionless environments. The architecture's performance, validated through multi-domain datasets, shows strong predictive capabilities with an ensemble F1 score of 0.88 and AUC exceeding 0.91, while maintaining minimal accuracy degradation under encrypted or federated settings. These results highlight that decentralized risk assessment can achieve high reliability without compromising user privacy or model integrity.

Despite these contributions, the study underscores several limitations that open pathways for future research. The increasing complexity of cross-chain user behavior necessitates expanded interoperability frameworks capable of harmonizing data across heterogeneous blockchain networks. Moreover, adversarial behaviors, such as coordinated transaction manipulation or synthetic identity creation, require more resilient anomaly- and fraud-detection modules enhanced through adversarial training or cryptographically verifiable behavioral proofs.

Future work should also focus on developing fully explainable encrypted models, enabling compliance with responsible AI standards while preserving the mathematical guarantees of privacy-preserving computation. Another promising direction is the design of decentralized governance systems that support continuous model updates through token-weighted voting, zk-proof-verified contributions, or autonomous risk oracles. Additionally, integrating decentralized identity (DID) and verifiable credentials may substantially improve creditworthiness estimation, particularly for users with limited on-chain histories.

Finally, large-scale real-world deployment requires alignment with emerging regulatory frameworks governing automated credit decisions, data sovereignty, and AI-driven financial risk management. As DeFi matures, secure AI-enhanced credit scoring will be pivotal in enabling fair, efficient, and globally accessible lending markets. The architecture proposed in this work provides a foundational blueprint for such next-generation decentralized credit systems, bridging the gap between advanced machine intelligence and the trustless nature of blockchain finance.

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